

XIN BARAG YOUQI, China

WMO: 506030

Lat: 48.6782N

Lon: 116.8100E

Elev: 1826

StdP: 13.75

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-20.7	-31.4	1.0	-24.0	-27.6	1.3	-19.9	26.3	4.2	22.5	2.2	5.1	250	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	91.7	64.6	88.1	63.5	84.5	62.6	69.1	81.5	67.6	79.5	66.3	77.8	9.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	101.1	73.1	64.0	95.4	72.7	62.2	89.7	71.8	34.5	81.1	33.2	79.5	32.1	77.6	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.6	21.2	18.1	-27.3	98.4	5.8	4.9	-31.5	102.0	-34.9	104.9	-38.1	107.7	-42.4	111.3		
			-27.6	71.7	5.7	1.6	-31.7	72.8	-35.0	73.8	-38.2	74.7	-42.4	75.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	-5.8	3.2	19.8	40.0	54.8	67.4	72.4	68.2	55.2	37.2	15.0	-1.2	(6)
(7)		DBStd	28.73	8.47	9.54	11.26	8.99	8.25	7.26	5.39	6.07	8.13	9.13	11.38	8.74	(7)
(8)		HDD50	7416	1729	1310	935	320	43	0	0	38	405	1049	1587		(8)
(9)		HDD65	11221	2194	1730	1400	750	331	55	6	35	307	862	1499	2052	(9)
(10)		CDD50	2198	0	0	0	20	193	524	695	563	195	8	0	0	(10)
(11)		CDD65	526	0	0	0	0	15	128	236	133	14	0	0	0	(11)
(12)		CDH74	5391	0	0	0	20	300	1454	2210	1204	202	1	0	0	(12)
(13)		CDH80	1973	0	0	0	3	94	571	843	406	56	0	0	0	(13)
(14)	Wind	WSAvg	7.8	6.9	7.5	8.5	9.8	9.8	7.6	6.8	7.1	7.7	7.9	7.1	6.8	(14)
(15)	Precipitation	PrecAvg	9.8	0.0	0.0	0.1	0.2	0.5	1.5	3.1	2.8	1.1	0.2	0.1	0.0	(15)
(16)		PrecMax	17.6	0.1	0.4	0.3	0.8	1.7	3.4	7.4	6.8	3.7	1.0	0.5	0.2	(16)
(17)		PrecMin	4.1	0.0	0.0	0.0	0.0	0.0	0.1	0.5	0.7	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.0	0.0	0.1	0.1	0.2	0.4	0.9	1.5	1.8	0.9	0.2	0.1	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.5	35.1	56.7	76.7	88.7	97.9	97.1	94.0	86.9	70.3	47.2	26.4	(19)
MCWB			18.2	27.0	40.6	50.6	57.1	63.0	67.1	65.2	59.9	51.4	36.6	22.7	(20)	
2%		DB	15.1	27.7	49.1	68.0	82.1	91.0	91.8	88.5	80.0	61.7	40.6	19.7	(21)	
		MCWB	12.8	22.6	36.1	46.1	54.7	62.3	66.5	64.0	57.8	45.8	32.6	17.0	(22)	
5%		DB	10.6	22.4	42.4	61.9	76.3	86.7	88.2	84.4	74.7	56.3	35.5	15.3	(23)	
		MCWB	8.6	18.8	32.0	43.3	52.9	61.5	65.9	63.6	56.1	43.0	29.3	13.2	(24)	
10%		DB	6.3	17.8	37.4	56.5	71.4	82.3	84.5	80.6	70.2	51.8	30.9	11.2	(25)	
		MCWB	4.8	15.0	29.3	40.6	50.5	60.2	64.7	63.0	54.5	40.8	26.3	9.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	27.6	41.1	51.0	59.8	67.6	72.0	69.8	64.5	51.3	37.1	22.8	(27)
MADB			21.2	34.5	56.3	75.2	81.1	82.5	84.7	82.2	74.2	69.1	46.6	26.0	(28)	
2%		WB	12.8	22.9	36.3	47.0	56.7	65.4	70.2	68.0	61.2	46.9	32.9	17.2	(29)	
		MADB	15.0	27.5	48.3	66.4	76.4	79.8	82.7	78.8	71.9	58.6	40.3	19.8	(30)	
5%		WB	8.7	18.9	32.6	43.8	54.3	63.9	68.6	66.8	58.7	44.0	29.6	13.2	(31)	
		MADB	10.6	22.4	41.7	60.5	72.5	79.4	81.0	77.6	71.0	54.7	35.1	15.3	(32)	
10%		WB	4.9	15.1	29.6	41.0	52.1	62.4	67.3	65.4	56.0	41.6	26.4	9.6	(33)	
		MADB	6.3	17.6	37.0	55.6	68.9	78.1	79.4	75.7	68.2	51.1	30.8	11.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.2	19.9	21.2	22.8	23.4	21.0	19.0	19.7	21.6	20.0	17.6	15.2	(35)
MADB			20.5	24.4	27.8	32.2	32.2	28.1	25.5	25.6	28.5	27.3	22.2	19.3	(36)	
MCWBR			18.6	20.2	17.8	16.7	14.1	10.0	8.2	8.9	12.6	15.6	15.5	17.0	(37)	
5% WB		MADB	20.4	24.3	26.5	30.4	28.0	21.6	19.6	18.9	22.6	24.8	21.7	19.3	(38)	
		MCWBR	18.5	20.0	17.4	16.1	12.8	8.1	7.6	8.0	11.0	14.7	15.4	17.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.260	0.283	0.327	0.403	0.485	0.472	0.434	0.394	0.346	0.319	0.285	0.267	(40)	
(41)		taud	2.250	2.211	2.179	2.068	1.880	2.042	2.226	2.306	2.374	2.353	2.274	2.205	(41)	
(42)		Ebn at Noon	256	275	279	266	248	252	261	267	271	259	241	232	(42)	
(43)		Edh at Noon	23	31	38	48	60	52	43	37	31	27	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	587	985	1426	1779	2035	2088	1947	1774	1428	992	611	454	(44)	
(45)		RadStd	35	53	83	89	100	143	114	119	85	51	30	35	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	-4.09	-5.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	-4.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page